

RODRIGO GOMES DE ARAÚJO

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EDUCATION

Master's

Computer Science

Sep 2025 – Present

University of Porto (FCUP)

Bachelor's

Computer Science and Engineering

Sep 2022 – Jul 2025

University of Porto (FCUP)

PROJECTS

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research on the state of the art of Bitcoin**, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing**, **Ed25519 signatures** and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **autonomous network behavior simulation** with clients, bootstrap nodes, well behaved nodes and malicious nodes.
- Developed for the Data Systems Security class and graded 19.1/20.

Self-Hosted Personal Portfolio | Next.js, React, TypeScript, Tailwind, Docker, Cloudflare, Tailscale

- **Personal portfolio** built with **Next.js and React in TypeScript**, styled with Tailwind and shadcn/ui.
- **Containerized with Docker** and published to Docker Hub, deployed and managed through **Portainer** on a personal home server.
- Exposed publicly via **Cloudflare Tunnel** with **no open inbound ports**, and administered remotely over a private **Tailscale** network.

Portuguese Elections Analytics Platform | PostgreSQL, PostGIS, Python, Plotly, FastAPI, OCaml

- **Interactive data visualization dashboard** for 10 years of Portuguese elections. Choropleth vote-swing maps, treemaps, radar and multi-year trend charts, drillable from country down to parish.
- **Geospatial rendering in-database**. Maps generated as SVG directly by PostGIS, with topology-preserving simplification adapted to zoom level.
- **ETL pipeline** into a **star-schema warehouse**, feeding **complex analytical queries** that join spatial and electoral data (recursive roll-ups, window functions, OLAP cubes).

RESEARCH

CaST (Cryptographically-aware Session Types)

Master's Thesis

Jul 2026 – Present

FreeST, Haskell, Type Systems, Applied Cryptography

- Extending the expressiveness of session types to capture **cryptographic resource usage**, so that its misuse in protocols is detected and rejected at compile-time.
- **Resource tracking composes across nested subprotocols** of arbitrary depth and is preserved under sequential composition and recursion.

AWARDS

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch

Oct 2025

- Rust and Raylib game from scratch in 48 hours, learning how to **reduce a problem to its core requirements**.

Academic Merit Awards

2020, 2021, 2022

Participation in RoboCup 2016 | RoboCup Federation

Jun 2016

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança

May 2016

TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL

Frameworks & Tools: Git, UNIX, Tokio, Docker, LibP2P, Dalek, Plotly, POSIX, STM, gRPC, PostgreSQL

Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)